$\bar{\nu}_e$ Appearance Analysis with the 3-Flavour Joint Fit Framework for T2K Run 1-9c (1.4938×10^{21}-POT Neutrino and 1.1236×10^{21}-POT Anti-Neutrino) Data Sets

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Abstract

We report the results of the $\bar{\nu}_e$ appearance analysis using the 3-flavour, 5 sample $\nu/\bar{\nu}$ analysis framework, performed by the *VALOR* group, on the combined Run 1-9c data set, using data taken in both neutrino and antineutrino beam mode. The goal of this analysis is to search for $\bar{\nu}_e$ appearance in a $\bar{\nu}_\mu$ beam and is an update to [13] The joint fit framework is used, whose details can be found in [14]. A joint analysis of all the Super-K samples, in both neutrino and antineutrino beam modes, is used in order to constrain the oscillation and systematic parameters, achieving a better sensitivity to $\bar{\nu}_e$ appearance. The Run 1-9c data set corresponds to an integrated J-PARC neutrino beam exposure of 1.4938×10^{21} POT in neutrino mode and 1.1236×10^{21} POT in antineutrino mode.

Using the Asimov A parameters (see Tab 2), this analysis predicts 268.5 ± 13.2 (syst) for neutrino μ-like, 95.5 ± 4.1 (syst) for antineutrino μ-like, 73.8 ± 7.1 (syst) for neutrino e-like, 11.8 ± 0.9 (syst) for antineutrino e-like and 6.9 ± 1.3 (syst) for neutrino ν_eCC1π^+-like events in Super K and observes respectively 243 for neutrino μ-like, 102 for antineutrino μ-like, 75 for neutrino e-like, 9 for antineutrino e-like and 15 for neutrino ν_eCC1π^+-like events.

A parameter β, that multiplies the $\bar{\nu}_\mu \to \bar{\nu}_e$ oscillation probability by either 0 or 1, is introduced in order to characterise the significance to $\bar{\nu}_e$ appearance. All the oscillation parameters, mass hierarchy included, are marginalised. The significance of the result is shown as p-values and a Bayes factor. Both a rate+shape and a rate-only analysis are performed.

For the Run 1-9c set and taking no $\bar{\nu}_e$ appearance as our null hypothesis we find a p-value of 0.209 ± 0.003 for the rate-only analysis and a p-value of 0.215 ± 0.003 for the rate+shape analysis. Taking nuebar appearance according to the best knowledge of the PMNS matrix as our null hypothesis we find a p-value of 0.210 ± 0.003 for the rate-only analysis and a p-value of 0.092 ± 0.003

for the rate+shape analysis. By throwing a large number of toys, both hypotheses account for uncertainties in the values of the oscillation parameters. The Bayes factor, given by the rate+shape analysis, was found to be 2.12 indicating a poor preference for the no $\bar{\nu}_e$ appearance hypothesis over the PMNS $\bar{\nu}_e$ appearance hypothesis.

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Parameter(s)	Prior	Range for marginalisation toys	Range for fake-data toys
$\sin^2\theta_{23}$	Uniform	$[0.3; 0.7]$	$[0.42; 0.62]$
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) reactors	Gaussian	0.0857 ± 0.0046	0.0857 ± 0.0046
$\sin^2 2\theta_{12}$	Gaussian	0.846 ± 0.021	0.846 ± 0.021
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{13}\rvert$ (IH)	Uniform	$[2.3; 2.7] \times 10^{-3}$ eV^2/c^4	$[2.30, 2.65] \times 10^{-3}$ eV^2/c^4
Δm^2_{21}	Gaussian	$(7.53 \pm 0.18) \times 10^{-5}$ eV^2/c^4	$(7.53 \pm 0.18) \times 10^{-5}$ eV^2/c^4
δ_{CP}	Uniform	$[-\pi; +\pi]$	$[-\pi, 1.1]\&[1.8, \pi]$
Mass Hierarchy	Bernoulli	0.5 normal, 0.5 inverted	0.5 normal, 0.5 inverted

Table 1: Treatment of the oscillation parameters in the $\bar{\nu}_e$ appearance analysis. All the gaussian priors are from [86]. The parameter $\sin^2 2\theta_{13}$ is constrained with the measurement of reactor experiments using the gaussian prior. The nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [86], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [23]. When generating fake-data toys used to fill the test-statistic distributions, the parameters that are thrown according to flat prior probability distributions have a restricted range compared to the priors used when generating marginalisation toys. This is purely to decrease the required CPU time when generating fake-data toys whilst having minimal impact on the p-values produced. This is detailed in appendix A.

1. Introduction

In this technical note, we report the results of the $\bar{\nu}_e$ appearance analysis using the 3-flavour $\nu/\bar{\nu}$ joint analysis, performed by the VALOR group, on the combined Run 1-9c data set, using data taken in both neutrino mode (1.4938×10^{21}), with the polarity set to forward horn current (FHC), and antineutrino (1.1236×10^{21}) mode with polarity set to reverse horn current (RHC). The same procedure used in the 2016 $\bar{\nu}_e$ appearance analysis [13] is used, whilst working in the analysis framework outlined in [14]. A parameter β, that multiplies by either 0 or 1 the $\bar{\nu}_\mu \to \bar{\nu}_e$ oscillation probability, is used in order to obtain the significance to $\bar{\nu}_e$ appearance. The priors used to marginalise the oscillation parameters and generate fake-data toys can be found in Tab 1.

The significance to $\bar{\nu}_e$ appearance is given as a p-value. Two different test statistics are used. The total number of observed events (N_{obs}) in the antineutrino mode e-like sample, without using any information from the reconstructed energy shape, is used to calculate the rate-only p-value. $\Delta\chi^2 = \chi^2(\beta = 0) - \chi^2(\beta = 1)$ using both rate and shape information from the five available samples (single-ring μ-like and single-ring e-like, both in FHC and RHC modes and FHC $\nu_e CC1\pi$-like) is used to calculate the rate+shape p-value. Where the χ^2 values are calculated by marginalising over all systematic and oscillation parameters, including the mass hierarchy (see section 1.3)

1.1. Improvements to the Previous Analysis

There have been a number of improvements since this analysis was previously performed, in 2016 [13]:

- The FHC exposure has increased by 113 % from 7.002×10^{20} to 1.4938×10^{21}
- The RHC exposure has increased by 50 % from 7.471×10^{20} to 1.1236×10^{21}
- An additional Super-K event sample has been added, neutrino mode ν_eCC1π-like

- The Super-K event reconstruction algorithm has changed from using APFit to FitQun [19]
- New cross-section model parameters are implemented, as detailed in [14]

1.2. Construction of Nominal Super-K Monte Carlo Templates

The Monte Carlo templates (predicted event spectra for each reaction mode produced by Monte Carlo simulation and binned in kinematic variables) used in this analysis were constructed in an analogous way to the procedure used for the 3-flavour joint analysis in [14]. The parameter of interest β which reweights the $\bar{\nu}_e$ appearance signal component is kept fixed to either 0 (corresponding to no appearance) or 1 (corresponding to appearance consistent with the current best knowledge of the PMNS matrix given by the simultaneous T2K+reactor Run 1-8 joint fit result) and is applied to all the Super-K samples, though its impact is negligible in all the samples except the antineutrino mode e-like sample.

In this analysis a comparison of the reconstructed energy (E_r) and reconstructed lepton angle (θ) spectrum of single ring events, including the ν_e CC1π^+-like sample, observed at Super-K with the predicted spectrum is used to build the test statistic to calculate the significance to $\bar{\nu}_e$ appearance. For μ-like samples the templates are built as a function of the reconstructed neutrino energy, as in previous analyses, and a single θ bin is used. The predicted number $N_{SK;r,\theta}$ of RHC single-ring e-like events in the r-th and θ-th bin is computed for each selected sample as follows:

$$N_{SK;r,\theta} = \sum_{m} \sum_{t} \sum_{r\prime} \left[\delta\left(m - m_{\bar{\nu}_\mu \to \bar{\nu}_e}\right)(\beta - 1) + 1\right] \cdot P_{m;t} \cdot T_{r;r\prime;f^{SK}_{E;r}} \cdot S_{m;t;r\prime,\theta;\vec{f}} \cdot N^{MC}_{SK;m;r\prime,\theta;t} \quad (1)$$

In equation 1, $N^{MC}_{SK;m;r\prime,\theta;t}$ is the input Super-K Monte Carlo (MC) template containing the number of events in the single ring MC sample with true reaction mode m in the true energy bin t, the reconstructed energy bin $r\prime$ and reconstructed lepton angle bin θ. $\delta\left(m - m_{\bar{\nu}_\mu \to \bar{\nu}_e}\right)$ is the Dirac delta function, equal to one when the reaction mode corresponds to the $\bar{\nu}_\mu \to \bar{\nu}_e$ oscillation and zero otherwise. $S_{m;t;r\prime,\theta;\vec{f}}$ is an overall, multiplicative, systematic error factor depending on a vector of nuisance (systematic) parameters $\vec{f}$, which is function of the reaction mode m, the true energy bin t, the reconstructed energy bin $r\prime$, and the reconstructed lepton angle bin θ. $T_{r;r\prime;f^{SK}_{E;r}}$ is a transfer function describing the migration of events between the reconstructed energy bins r and $r\prime$ due to uncertainty in the Super-K reconstructed energy scale, expressed here in terms of the nuisance parameter $f^{SK}_{E;r}$. Finally, $P_{m;t}$ is the 3-flavour oscillation probability applied to the true energy bin t of the Super-K MC template which corresponds to mode m; This is exactly how the analysis templates are built in the joint fit analysis, with the only difference that the $\bar{\nu}_\mu \to \bar{\nu}_e$ component can be switched on or off by the parameter β in each Super-K sample. In Fig 3 the predicted spectra are compared with the observed data. Also the 2-dimensional spectrum of the $\bar{\nu}_\mu \to \bar{\nu}_e$ component, that can be switched on or off by the parameter β is shown.

1.3. Calculation of the Marginal Likelihood

In the analyses presented here both the likelihood and the marginal likelihood are used. First the log-likelihood function is calculated for each sample, s as follows:

$$-2 \ln \lambda_s(\beta, \mathbf{a}) = 2 \cdot \sum_{i=0}^{N-1} \left(n_{s,i}^{obs} \cdot \ln(n_{s,i}^{obs}/n_{s,i}^{exp}) + (n_{s,i}^{exp} - n_{s,i}^{obs}) \right) \tag{2}$$

where $n_{s,i}^{obs}$ is the observed number of events in the ith bin in reconstructed energy and lepton angle of sample s and $n_{s,i}^{exp} = n_{s,i}^{exp}(\beta; \mathbf{a})$ is the corresponding expected number of events, N is the number of reconstructed energy and lepton angle bins and $\mathbf{a}$ is a vector of systematic parameters.

The total likelihood is then given by the sum of $-2 \ln \lambda_s(\beta, \mathbf{a})$ from all of the Super-K samples

$$-2 \ln \lambda(\beta, \mathbf{a}) = \sum_{s=1}^{N_{samples}} -2 \ln \lambda_s(\beta, \mathbf{a}) \tag{3}$$

where $N_{samples}$ corresponds to the number of Super-K samples used in the analysis. The marginal likelihood is then computed by considering an ensemble of MC predictions, generated based on priors on nuisance parameters, as in [14], and priors on oscillation parameters from Tab 1.

$$\chi^2_{marg}(\beta) = -2 \ln \lambda(\beta) = -2 \ln \left(\frac{1}{N_{marg}} \sum_{j=1}^{N_{marg}} \lambda(\beta, \mathbf{a}_j) \right) \tag{4}$$

It is important to note that this numerical integration is a weighted average of the likelihood evaluated at N_{marg} throws of all oscillation and systematic parameters, not a weighted average of the log-likelihood.

2. $\bar{\nu}_e$ Appearance Methodology

The significance of $\bar{\nu}_e$ appearance is given as a p-value by introducing the parameter β in order to compare the data to two null hypotheses: a) No $\bar{\nu}_e$ appearance ($\beta = 0$) and b) $\bar{\nu}_e$ appearance according to the current best knowledge of the PMNS matrix ($\beta = 1$), where both hypotheses fully account for uncertainties in the values of the oscillation parameters. Two different test statistics are used: rate-only and rate+shape.

An ensemble of toys are generated for each null hypothesis, randomising nuisance parameters according to the priors in Tab 1 and applying statistical fluctuations. In order to constrain the oscillation and systematic model parameters using T2K data from the "control samples" (neutrino mode single-ring and multi-ring e-like and both neutrino and antineutrino mode single-ring μ-like), the toy distributions are weighted according to the following method, which ensures that the test-statistic distributions are filled with only the most likely toys and maintains correlations between all nuisance parameters in the 4 "control samples":

1. A large number ($> 1M$) of toy experiments are produced taking random parameters from the distributions in Tab 1 and with statistical fluctuations applied.

2. The 4 sample (i.e. excluding antineutrino mode single-ring e-like) likelihood of the toy given the data is calculated for each toy using equation 5. Then the maximum likelihood, λ_{max}, is stored.
3. A toy experiment is produced taking random parameters from the distributions in Tab 1 to form the proposed distribution, $T_{proposed}$.
4. The 4 sample likelihood of $T_{proposed}$ is found: $\lambda_{proposed}$.
5. A random number R is drawn uniformly from $\{0:1\}$.
6. If $\frac{\lambda_{proposed}}{\lambda_{max}}$ is greater than R, $T_{proposed}$ is accepted and used in the distribution.
7. The procedure is repeated from step 3 until 20k of $T_{proposed}$ toys have been accepted, denoted $T_{accepted}$. The resulting set of $T_{accepted}$ toys are constrained by the likelihood of the "control samples" given the data and only contain the most likely toys, so are referred to as weighted toys
8. The rate-only and rate+shape test-statistics are calculated according to sections 2.1 and 2.2 for each $T_{accepted}$ to produce test-statistic distributions.

$$\lambda\left(Toy; Data\right) = \prod_{s=1}^{N_{samples}=4} \lambda_s\left(Toy; Data\right) \tag{5}$$

It is worth noting that this method of weighting the toy distributions is statistically equivalent to performing a likelihood ratio fit of the "control samples" and using the posterior distribution of the nuisance parameters as a prior to extract the systematic parameters for a toy experiment.

At this point, the rate-only and rate+shape analysis methods diverge.

2.1. Rate-only Analysis Method

The total number of observed events, N_{obs}, in the antineutrino mode e-like sample, without using any information from the reconstructed energy and angle distribution, is used to quote the rate-only p-value.

The expected distribution of antineutrino mode single-ring e-like events, under statistical and systematic fluctuations, is produced using 20k (throws of nuisance parameters) x 10k (throws of statistical fluctuations) toy data sets to ensure the nuisance parameter space was scanned effectively. In this method, the toy experiments used for the 20k nuisance parameter throws are the sampled toys described in section 2. For each of these toys, 10k statistical throws are performed using a Poisson distribution with mean equal to the number of antineutrino mode single-ring e-like events in the toy. The result of each statistical throw, N_{exp}, is stored in the rate-only test-statistic distribution, $f\left(N_{exp}\right)$.

After performing statistical throws, a p-value is calculated for $f\left(N_{exp}\right)$. For the $\beta = 0$ null hypothesis, a right-tailed p-value is used (see equation 6), while for the $\beta = 1$ null hypothesis, a left-tailed p-value is used (see equation 7). A right-tailed p-value is used for the $\beta = 0$ hypothesis as the $\beta = 0$ test-statistic distribution is composed solely of background events. The p-value is the probability of obtaining a result as extreme or more extreme than the observed result under the null

hypothesis, so using a right-tailed p-value gives the probability of observing a fluctuation under the $\beta = 0$ hypothesis in the background with equal to or more events than observed in the antineutrino mode single-ring e-like sample. An upwards fluctuation such as this would be less consistent with coming from the expected number of events from a sample containing only background events. Therefore, if this p-value is lower than some threshold (usually set to $5\,\sigma$), it could be claimed that the observation is incompatible with having come from the null hypothesis and the hypothesis is rejected. Similarly, a left-tailed p-value is used for the $\beta = 1$ hypothesis as the $\beta = 1$ test-statistic distribution is composed of both background and signal events, so the left-tailed p-value gives the probability of observing a fluctuation under the $\beta = 1$ hypothesis with equal to or fewer events than were observed in the antineutrino mode single-ring e-like sample. A downwards fluctuation such as this would be less consistent with coming from the expected number of events from a sample containing both background and signal events.

$$\text{p-value(right tail)} = \frac{\int_{N_{obs}}^{+\infty} f\left(N_{exp}\right) dN_{exp}}{\int_{-\infty}^{+\infty} f\left(N_{exp}\right) dN_{exp}} \tag{6}$$

$$\text{p-value(left tail)} = \frac{\int_{-\infty}^{N_{obs}} f\left(N_{exp}\right) dN_{exp}}{\int_{-\infty}^{+\infty} f\left(N_{exp}\right) dN_{exp}} \tag{7}$$

As the distribution of the statistic is populated with such a large sample, the statistical uncertainty on the p-value is very small, of the order of 10^{-3}.

2.2. Rate+shape Analysis Method

The rate+shape analysis method uses log-likelihood ratio between the $\beta = 0$ and $\beta = 1$ $\bar{\nu}_e$ appearance null hypotheses as the test-statistic:

$$\Delta\chi^2 = \chi^2_{marg}\left(\beta = 0\right) - \chi^2_{marg}\left(\beta = 1\right) \tag{8}$$

This test statistic simply compares the oscillation with the non-oscillation hypothesis but, unlike the $\Delta\chi^2$ values given in the 3-flavour joint analysis [14], this does not use the best-fit χ^2 value, as β is not fit for specifically. For this reason the distribution of $\Delta\chi^2$ is expected to take both negative and positive values, where $\Delta\chi^2 < 0$ (> 0) corresponds to the spectra being more consistent with the $\beta = 0$ (1) hypothesis than $\beta = 1$ (0). It is worth noting that the parameter β is not fit, but is fixed either to 0 or 1. It is also worth noting that marginal χ^2 is calculated by using all 5 Super-K samples.

The $\Delta\chi^2$ distribution is produced by first generating 20k toys produced using the method described in section 2. The toys are then marginalised using 200k marginalisation toys generated from the priors in Tab 1 and $\chi^2_{marg}\left(\beta\right)$ is calculated according to equation 4. This incorporates the uncertainty on nuisance parameters on the measurement and distributions of the test-statistic.

Once a $\Delta\chi^2$ value is calculated for each toy, it is stored in the test-statistic distribution, $f\left(\Delta\chi^2\right)$, which is integrated to obtain p-values, for example the right-tailed p-value for the $\beta = 0$ null hypothesis shown in equation 9.

$$\text{p-value(right tail)} = \frac{\int_{\Delta\chi^2_{obs}}^{+\infty} f\left(\Delta\chi^2\right) d\Delta\chi^2}{\int_{-\infty}^{+\infty} f\left(\Delta\chi^2\right) d\Delta\chi^2} \tag{9}$$

The number of statistical throws from each toy data set is different between the rate-only and rate+shape analyses. The rate+shape analysis only uses a single statistical fluctuation upon generating each fake-data toy. The reason for this is because the limiting factor in the rate+shape analysis is the calculation of the marginal likelihood. 200k marginalisation toys were used to determine the marginal likelihood in order to explore the whole space of the systematic parameters. More marginalisation toys are used compared to the joint fit analysis, because in this analysis the mass hierarchy, δ_{CP} and all other oscillation parameters are marginalised at the same time whereas they were not in the joint analysis. Furthermore these parameters have a large impact on the $\nu_\mu \rightarrow \nu_e$ component which is a significant background in the null hypotheses.

It is worth pointing out that sometimes the rate-only p-value can be smaller than the rate+shape. This can be due to both the discreteness of the rate-only p-value or to the shape information (which the rate-only analysis does not have access to) which can discredit the $\bar{\nu}_e$ appearance hypothesis. For this reason it is important to quote both the p-values as our sensitivity.

Both the rate-only and rate+shape analyses were validated against what was found by the P-theta group. For the rate-only analysis we see good agreement between the two analyses, as expected. For the rate+shape analysis, the only difference between P-theta and VALOR is due to the e-like samples. For VALOR the e-like samples are binned in $E_{reco} - \theta$ analysis templates whereas for P-theta they are binned in $p - \theta$. When the P-theta group runs the analysis using $E_{reco} - \theta$, the results of VALOR and P-theta agree reasonably well. In addition, the P-theta group performed fake-data studies to assess the impact of uncertainties in the neutrino-nucleus interaction model and found no evidence of any significant bias [20].

2.3. Statistical Uncertainty on the p-values

The statistical uncertainty on the p-value is calculated assuming that the number of entries in each bin of the expected distribution of the test statistic follow a Poisson distribution. The p-value uncertainty is then derived from the statistical uncertainty of the population. As the p-value is calculated from the fraction of trials which fall to the right of the data statistic, the p-value is expected to be binomially distributed. Given a sufficiently large number of trials (N), the p-value is expected to take a gaussian distribution with a mean of the p-value and a standard deviation of $\sqrt{p(1-p)/N}$.

The uncertainty is calculated from the first order Taylor series expansion assuming no correlation between bins in the test-statistic distribution and the number of events in each bin follows a Poisson distribution. The calculation of the uncertainty summing over all toy experiments, for the right sided rate-only p-value, is shown as an example and is as follows:

The p-value calculation shown in equation 6 is written in terms of an integral over the test-statistic distribution. This is re-written as a sum of all bins that contribute to the right-tail of the distribution

$$\text{p-value(right tail)} = p = \frac{\int_{N_{obs}}^{+\infty} f\left(N_{exp}\right) dN_{exp}}{\int_{-\infty}^{+\infty} f\left(N_{exp}\right) dN_{exp}} = \frac{\sum_{i=0}^{n(right-tail)} n_i}{\sum_{i=0}^{n(total)} n_i} \tag{10}$$

Where n_i is the the number of times each toy, i, appears. The total uncertainty as a result of the statistical uncertainty on each sample is given by

$$(\sigma_p)^2 = \sum_i \left(\frac{\partial p}{\partial n_i} \sigma_{n_i} \right) \tag{11}$$

Expanding and using the Poisson uncertainty on the toys $\sigma_{n_i} = \sqrt{n_i}$ before taking the limit where each toy appears only once gives

$$(\sigma_p)^2 = \frac{p(1-p)}{N_{tot}} \tag{12}$$

Where N_{tot} is the integral of the test-statistic distribution. This is the expected result for a binomial process with a success rate p.

2.4. Bayes Factor

As in this analysis the $\Delta\chi^2$ statistic considered is marginalised over all nuisance parameters, the statistic can be interpreted in a Bayesian framework as a Bayes factor. The Bayes factor is another statistic (fully defined by the rate+shape analysis $\Delta\chi^2$ statistic) and gives an indication of which hypothesis gives the best fit to the data. The Bayes factor is defined as the marginal likelihood ratio of the two hypotheses in question.

$$B_{01} = \frac{e^{-\chi^2_{\beta=0}/2}}{e^{-\chi^2_{\beta=1}/2}} = e^{-\frac{1}{2}\Delta\chi^2} \tag{13}$$

The Jeffreys scale [78] is used to interpret the Bayes factor to show whether there is a preference for one hypothesis over the other. A more detailed discussion of Bayes factors can be found in [79].

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	175.69877	0.00417	11.13658	0.00020	0.02258	0.00015	186.86245
CC1pi	27.64351	0.00201	2.64220	0.00009	0.02418	0.00007	30.31206
CCcoherent	0.28884	0.00000	0.09306	0.00000	0.00000	0.00000	0.38190
2p2h	35.49529	0.00154	1.42670	0.00008	0.02578	0.00004	36.94944
CCother	5.49849	0.00087	0.41118	0.00008	0.00011	0.00001	5.91073
NC1pi0	0.61317	0.01856	0.02142	0.00145	N/A	N/A	0.65460
NC1piPM	4.89912	0.09882	0.17708	0.00990	N/A	N/A	5.18492
NCcoherent	0.00000	0.00000	0.00044	0.00004	N/A	N/A	0.00047
NCother	2.06264	0.07441	0.13205	0.00926	N/A	N/A	2.27837
Total	252.19982	0.20038	16.04070	0.02110	0.07266	0.00027	268.53493

Table 3: Predicted number of events in the neutrino mode single-ring μ-like sample obtained after applying the BANFF weight for a total exposure of 1.4938×10^{21} FHC POT. The Asimov data set A in Tab 2 is used. The total of 268.53493 events consists of 252.19982 signal events and 16.33511 background events.

3. Sensitivity to $\bar{\nu}_e$ Appearance

Since there are no fitted parameters in the analysis, the validation can be performed by checking for agreement between the oscillation analyses from both the VALOR and P-theta groups. This is done by performing a sensitivity study to $\bar{\nu}_e$ appearance data given the Asimov A data set, which is treated as we would treat the data. The Asimov A data set consists of the most probable data set (obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [23]) that is the nominal expected E_{reco} vs θ distribution [68] and the corresponding parameters are listed in Tab 2. The expected event rates for all five Super-K event samples using Asimov A parameters are shown in Tables 3 to 7.

Parameter(s)	Asimov A
$\sin^2\theta_{23}$	0.528
$\sin^2\theta_{13}$ reactors	0.0219
$\sin^2\theta_{12}$	0.304
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{31}\rvert$ (IH)	2.509×10^{-3} eV^2/c^4
Δm^2_{21}	7.53×10^{-5} eV^2/c^4
δ_{CP}	-1.601
Mass Hierarchy	Normal
β	1

Table 2: Values of oscillation parameters used to compute the event rates, systematic effects and sensitivity studies corresponding to the Asimov A data set, which is the MC expected distribution in a certain oscillation hypothesis. In the Asimov data set A the nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [86], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [23]. The mass hierarchy is not marginalised but fixed to either NH or IH.

The significance to $\bar{\nu}_e$ appearance is quoted by producing p-values using the procedure described in Sec 2. The expected Run 1-9c statistics sensitivity studies were performed for 1.4938×10^{21} POT and 1.1236×10^{21} POT in neutrino mode and antineutrino mode respectively. The distributions of the event-rate in the antineutrino mode single-ring e-like sample for both hypotheses are shown in

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.15943	6.36106	0.00638	0.25944	46.48200	0.28047	53.54879
CC1pi	0.04653	0.93873	0.00282	0.06851	4.48395	0.04775	5.58828
CCcoherent	0.00011	0.00836	0.00018	0.00403	0.03441	0.00330	0.05039
2p2h	0.02985	1.60917	0.00118	0.04339	8.69070	0.03920	10.41351
CCother	0.01337	0.12912	0.00035	0.00957	0.08761	0.00328	0.24329
NC1pi0	1.69779	0.03737	0.06073	0.00362	N/A	N/A	1.79950
NC1piPM	0.16808	0.00476	0.01032	0.00058	N/A	N/A	0.18374
NCcoherent	0.52906	0.00721	0.04837	0.00298	N/A	N/A	0.58763
NCother	0.31903	0.01710	0.02145	0.00130	N/A	N/A	0.35888
NC1gamma	0.94061	0.01701	0.05029	0.00237	N/A	N/A	1.01028
Total	3.90385	9.12991	0.20207	0.39579	59.77868	0.37400	73.78429

Table 4: Predicted number of events in the neutrino mode single-ring e-like sample obtained after applying the BANFF weight for a total exposure of 1.4938×10^{21} FHC POT. The Asimov data set A in Tab 2 is used. The total of 73.78429 events consists of 59.77868 signal events and 14.00562 background events.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	22.71950	0.00126	44.41840	0.00138	0.00140	0.00204	67.14397
CC1pi	5.87728	0.00050	6.74025	0.00037	0.00089	0.00091	12.62020
CCcoherent	0.06995	0.00000	0.26113	0.00007	0.00000	0.00009	0.33125
2p2h	6.64836	0.00038	4.23289	0.00040	0.00065	0.00031	10.88300
CCother	1.40166	0.00018	0.90187	0.00000	0.00000	0.00000	2.30370
NC1pi0	0.08919	0.00330	0.09127	0.00251	N/A	N/A	0.18627
NC1piPM	0.53424	0.02270	0.68366	0.01835	N/A	N/A	1.25895
NCcoherent	0.00000	0.00000	0.00299	0.00000	N/A	N/A	0.00299
NCother	0.46952	0.02440	0.29873	0.01173	N/A	N/A	0.80438
Total	37.80970	0.05271	57.63119	0.03482	0.00294	0.00335	95.53471

Table 5: Predicted number of events in the antineutrino mode single-ring μ-like sample obtained after applying the BANFF weight for a total exposure of 1.1236×10^{21} RHC POT. The Asimov data set A in Tab 2 is used. The total of 95.53471 events consists of 57.63119 signal events and 37.90353 background events.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.01912	0.75719	0.03547	1.25946	1.52367	4.23540	7.83030
CC1pi	0.01132	0.15029	0.01089	0.21446	0.21281	0.49728	1.09705
CCcoherent	0.00004	0.00148	0.00062	0.01585	0.00169	0.04650	0.06618
2p2h	0.00437	0.21366	0.00326	0.16792	0.34562	0.49049	1.22533
CCother	0.00578	0.03146	0.00178	0.01716	0.01547	0.00993	0.08158
NC1pi0	0.20578	0.00804	0.26242	0.00695	N/A	N/A	0.48317
NC1piPM	0.03677	0.00189	0.02971	0.00120	N/A	N/A	0.06958
NCcoherent	0.07106	0.00329	0.30096	0.00472	N/A	N/A	0.38002
NCother	0.08422	0.00327	0.03969	0.00147	N/A	N/A	0.12865
NC1gamma	0.11798	0.00653	0.25971	0.00570	N/A	N/A	0.38992
Total	0.55643	1.17710	0.94452	1.69488	2.09927	5.27959	11.75180

Table 6: Predicted number of events in the antineutrino mode single-ring e-like sample obtained after applying the BANFF weight for a total exposure of 1.1236×10^{21} RHC POT. The Asimov data set A in Tab 2 is used. The total of 11.75180 events consists of 5.27959 signal events and 6.47220 background events.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.02845	0.02467	0.00118	0.00177	0.21197	0.00268	0.27071
CC1pi	0.07487	0.76673	0.00258	0.00389	4.85230	0.00229	5.70266
CCcoherent	0.00026	0.01781	0.00008	0.00007	0.11156	0.00011	0.12989
2p2h	0.01000	0.02214	0.00049	0.00096	0.15884	0.00106	0.19349
CCother	0.03763	0.11311	0.00117	0.00284	0.11952	0.00151	0.27577
NC1pi0	0.02039	0.00076	0.00084	0.00008	N/A	N/A	0.02208
NC1piPM	0.06503	0.00166	0.00386	0.00033	N/A	N/A	0.07089
NCother	0.20699	0.00838	0.01319	0.00090	N/A	N/A	0.22945
NC1gamma	0.02853	0.00000	0.00068	0.00007	N/A	N/A	0.02928
Total	0.47215	0.95526	0.02408	0.01091	5.45418	0.00764	6.92422

Table 7: Predicted number of events in the neutrino mode ν_eCC1π^+-like sample obtained after applying the BANFF weight for a total exposure of 1.4938×10^{21} FHC POT. The Asimov data set A in Tab 2 is used. The total of 6.92422 events consists of 5.45418 signal events and 1.47004 background events.

β	Analysis	p-value	σ	Asimov A Expectation
0	rate-only	0.036 ± 0.001	2.07	N.A.
	rate+shape	0.0128 ± 0.0008	2.49	
1	rate-only	0.386 ± 0.003	0.87	$N_{events} = 11.8 \pm 0.9$
	rate+shape	0.426 ± 0.003	0.80	$\Delta\chi^2 = 4.592$

Table 8: Asimov A p-values and event-rates for the $\beta = 0$ and $\beta = 1$ hypotheses using both the rate-only and rate+shape analyses. There are no Asimov A expectations for the $\beta = 0$ hypothesis, as the Asimov A dataset assumes that $\beta = 1$ is true.

Fig 1 and the $\Delta\chi^2$ distributions for both hypotheses are shown in Fig 2. The corresponding Asimov A p-values are shown in Tab 8.

From the study of the Asimov data set A, which expects approximately 5 signal and 6 background events in the antineutrino mode e-like sample, we expect a p-value of about $1 - 2\,\%$ for the $\beta = 0$ hypothesis, corresponding to an exclusion of about $2.3 - 2.6\,\sigma$, where the conversion to σ is done using equation 14. We find a very large p-value of about $30\,\%$ in both analyses for the $\beta = 1$ hypothesis, corresponding to an exclusion of $< 1\,\sigma$, which is expected as Asimov data set A was generated using the $\beta = 1$ hypothesis. This is supported by the Bayes factor, which was found to be 0.101, indicating a substantial preference for the $\beta = 1$ hypothesis over the $\beta = 0$ hypothesis. The P-theta rate-only p-values for the Asimov A sensitivity study are 0.0383 and 0.373 for $\beta = 0$ and $\beta = 1$, respectively and the rate+shape p-values (using $E_{reco} - \theta$) are 0.0147 and 0.407 for $\beta = 0$ and $\beta = 1$, respectively. The p-values produced by the VALOR and P-theta groups are considered to be within acceptable agreement. The reason for the remaining differences between the VALOR and P-theta p-values are thought to be due to differences in the toy generation methods.

$$\sigma = \sqrt{2}\,\mathrm{erf}^{-1}\,(1 - p) \tag{14}$$

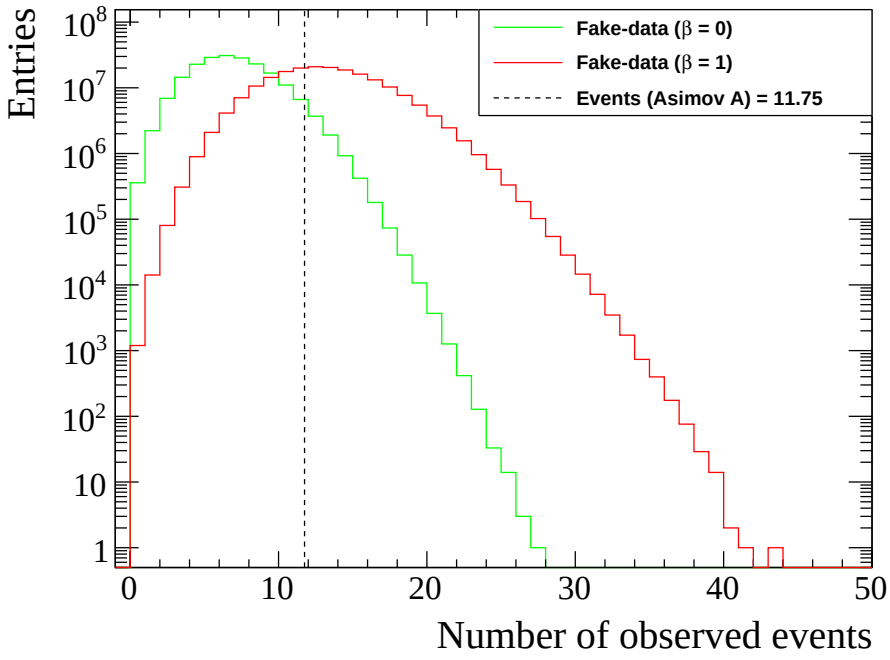


Figure 1: The distribution of the total number of events in the antineutrino mode single-ring e-like sample from each of the 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses. The distributions were produced using the method outlined in section 2.1 using the Asimov A spectra to calculate the 4-sample likelihood. The dashed line corresponds to the total number of events given by the Asimov A data set.

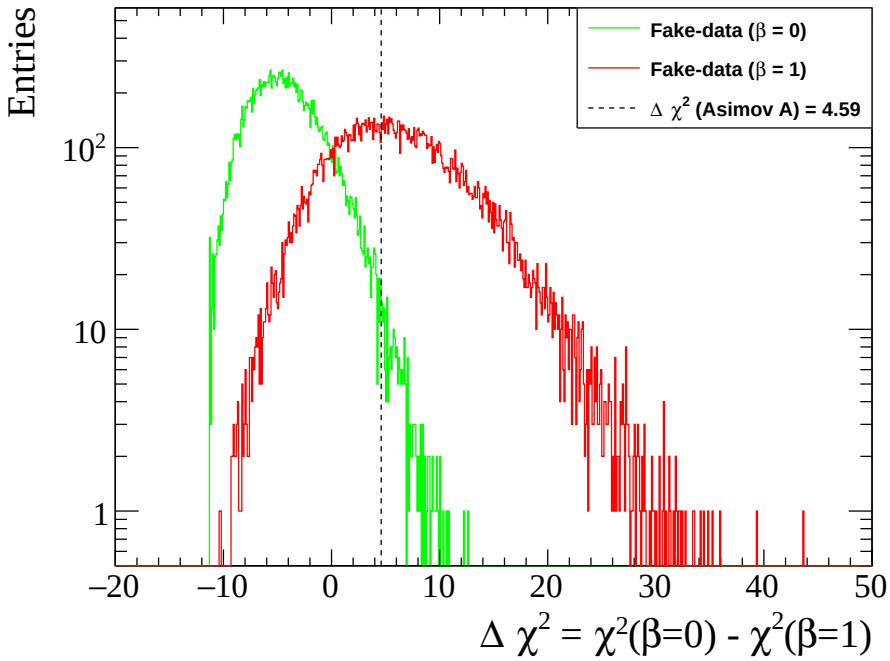
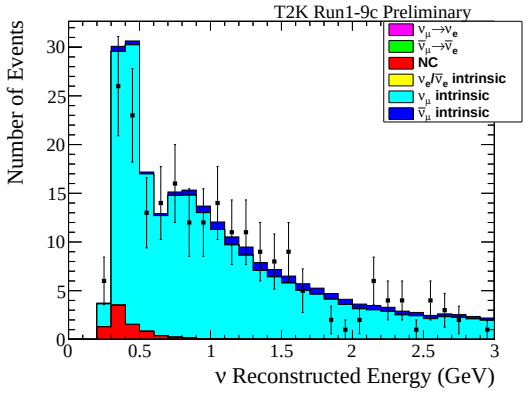


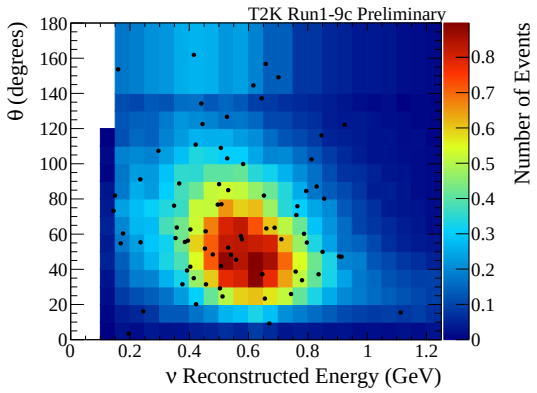
Figure 2: The distribution of $\Delta\chi^2$ values from each of the 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses. The distributions were produced using the method outlined in section 2.2 using the Asimov A spectra to calculate the 4-sample likelihood. The dashed line corresponds to the $\Delta\chi^2$ value given by the Asimov A data set.

4. Results of the $\bar{\nu}_e$ Appearance Analysis

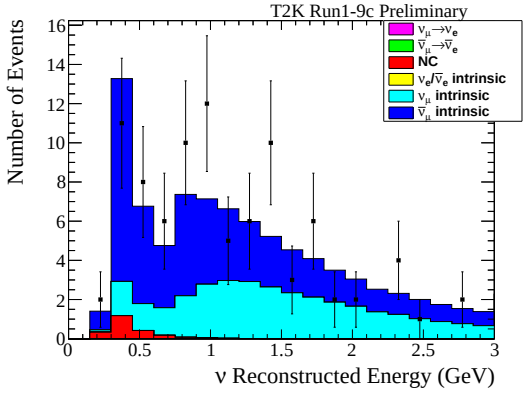
The rate-only and rate+shape p-values for $\bar{\nu}_e$ appearance are performed on the combined Run 1-9c statistics following the same procedure as in Sec 2. 1-sided p-values are computed on the right tail of the test statistic expected distribution to exclude $\beta = 0$ and the left tail to exclude $\beta = 1$. Fig 3 shows the data in the 5 Super-K event samples overlaid with the expected event rates for Asimov data set A. A summary of the observed number of events in Super-K, $\Delta\chi^2$ and p-values is given in Tab 9.



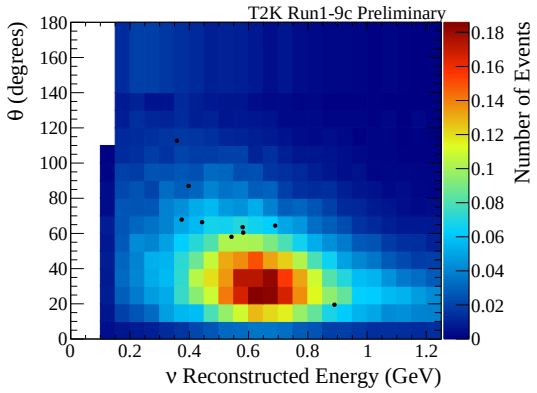
(a) Muon-like, neutrino mode



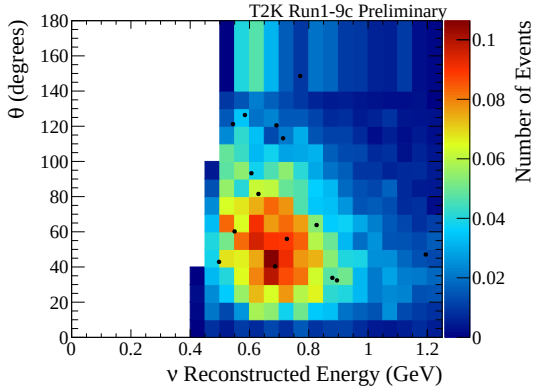
(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode



(d) Electron-like antineutrino mode



(e) ν_e CC1π^+-like, neutrino mode

Figure 3: Predicted Asimov A spectra. μ-like distributions are a function of the reconstructed neutrino energy, while the e-like, including ν_e CC1π^+, distributions are functions of both the reconstructed neutrino energy and the reconstructed angle between the outgoing lepton and the neutrino direction. The distributions correspond to the statistics collected in the full Run 1-9c data set. The spectra are generated with the oscillation parameters corresponding to the Asimov data set A shown in table 2.

Run	RHC POT	Rate-only Analysis			Rate+shape Analysis			
		N_{obs}	$\beta = 0$ p-value	$\beta = 1$ p-value	$\Delta\chi^2_{\text{obs}}$	$\beta = 0$ p-value	$\beta = 1$ p-value	B_{01}
1-9c	1.1236×10^{21}	9	0.209 ± 0.003	0.210 ± 0.003	-1.502	0.215 ± 0.003	0.092 ± 0.003	2.12
1-7	7.471×10^{20}	4	0.410	0.2155	-1.930	0.3742	0.099	2.62

Table 9: Summary of observed number of events, $\Delta\chi^2$, rate-only and rate+shape p-values and Bayes factor from the combined Run 1-9c data set compared to the Run 1-7 data set used for the previous iteration of this analysis, in 2016.

In Figs 4 and 5 the distributions of the test statistic, for rate-only and rate+shape analyses, are shown and compared with the data. Tab 9 contains a summary of the results. The rate-only analysis disfavours each hypothesis almost equally, whilst the rate+shape analysis disfavours the $\beta = 1$ hypothesis more than the $\beta = 0$ hypothesis. The Jeffreys scale indicates that the Bayes factor of 2.12 corresponds to a poor preference for the $\beta = 0$ hypothesis over $\beta = 1$. In comparison to the Run 1-7 analysis, the rate-only $\beta = 0$ p-value has decreased by a factor of 0.51. This is due to the number of observed events per POT increasing by a factor of 1.5, increasing the amount that the $\beta = 0$ hypothesis is disfavoured. The rate+shape $\beta = 0$ p-value has decreased by a factor of 0.57. In addition to the number of observed events per POT increasing by a factor of 1.5, this can be explained by an increase in the number of events (compared to the Run 1-7 analysis [13]) in regions of the spectra in Fig 3d where a larger number of events are expected.

The P-theta rate-only p-values for the Run 1-9c data set are 0.219 and 0.213 for $\beta = 0$ and $\beta = 1$, respectively and the rate+shape p-values (using $E_{reco} - \theta$) are 0.214 and 0.0843 for $\beta = 0$ and $\beta = 1$, respectively. The p-values produced by the VALOR and P-theta groups are considered to be within acceptable agreement. The reason for the remaining differences between the VALOR and P-theta p-values are thought to be due to differences in the toy generation methods.

By breaking the electron-like antineutrino mode spectra from Fig 3 into its components, as shown in Fig 6, some explanation can be offered for the preference for the no $\bar{\nu}_e$ appearance hypothesis. The observed events do not appear to fit the Asimov A expected $\bar{\nu}_\mu \to \bar{\nu}_e$ distribution well, instead they are more consistent with the $\nu_\mu \to \nu_e$ and intrinsic ν_e & $\bar{\nu}_e$ backgrounds. As the shape of the spectra discredits the $\beta = 1$ hypothesis (which is information that the rate-only analysis does not have access to), the $\beta = 1$ p-value for the rate+shape analysis is substantially smaller than that of the rate-only analysis.

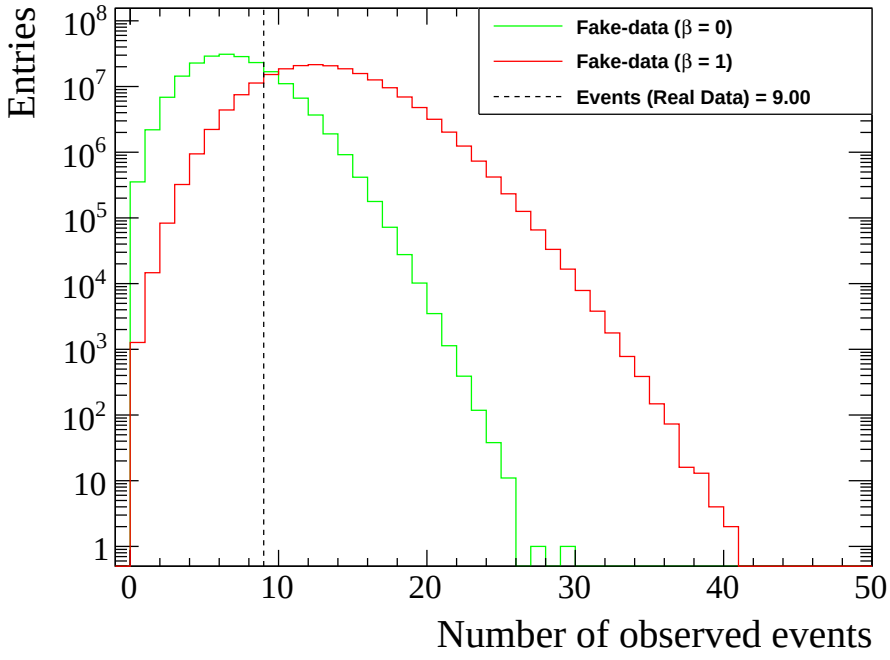


Figure 4: The distribution of the total number of events in the antineutrino mode single-ring e-like sample from each of the 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses. The distributions were produced using the method outlined in section 2.1 using the Run 1-9c data spectra to calculate the 4-sample likelihood. The dashed line corresponds to the total number of events given by the Run 1-9c data set.

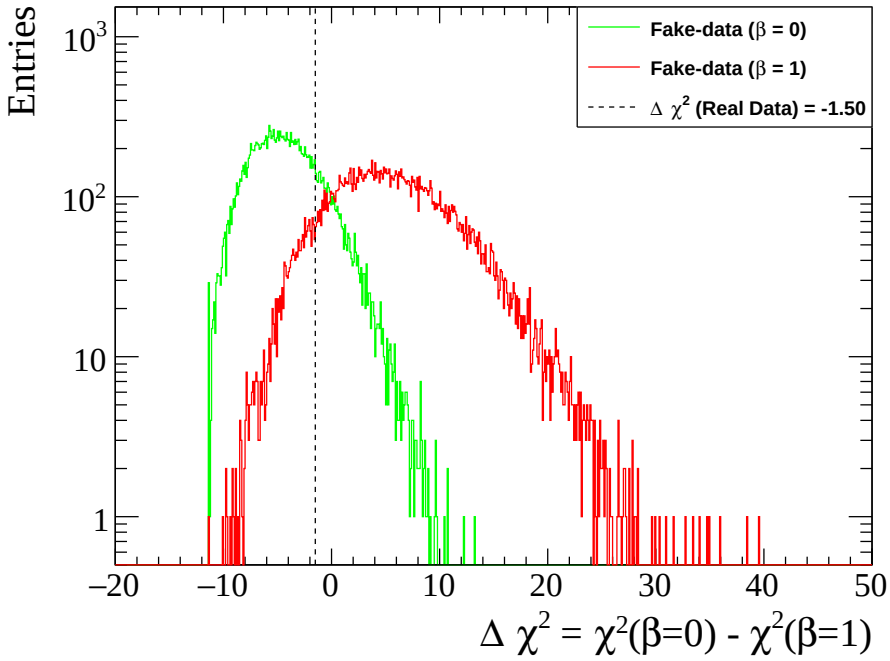
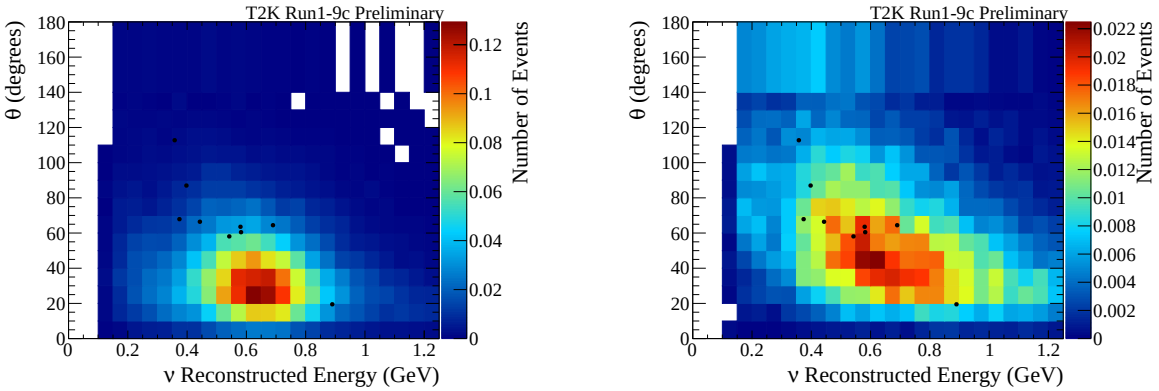
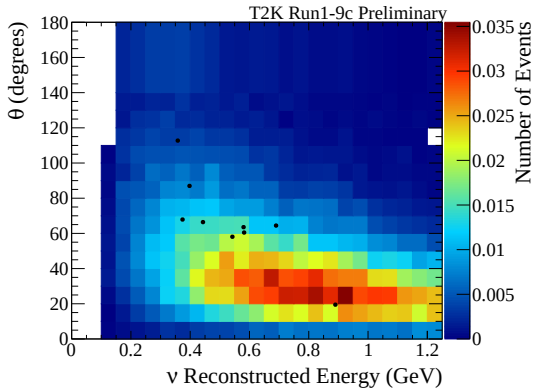


Figure 5: The distribution of $\Delta\chi^2$ values from each of the 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses. The distributions were produced using the method outlined in section 2.2 using the Run 1-9c data spectra to calculate the 4-sample likelihood. The dashed line corresponds to the $\Delta\chi^2$ value given by the Run 1-9c data set.



(a) $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$

(b) $\nu_\mu \rightarrow \nu_e$



(c) Intrinsic ν_e + Intrinsic $\bar{\nu}_e$

Figure 6: Predicted Asimov A Electron-like antineutrino mode spectra broken down by component. The distributions correspond to the statistics collected in the full Run 1-9c data set. The spectra are generated with the oscillation parameters corresponding to the Asimov data set A shown in table 2.

5. Summary

In this technical note, we have presented the results of the $\bar{\nu}_e$ appearance analysis using the 3-flavour $\nu/\bar{\nu}$ joint analysis framework performed by the VALOR group, on the combined Run 1-9c data set, for an exposure of 1.4938×10^{21} POT in neutrino mode and 1.1236×10^{21} POT in antineutrino mode. The goal of the analysis is to search for $\bar{\nu}_e$ appearance in a $\bar{\nu}_\mu$ beam.

This analysis predicts 268.5 ± 13.2 (syst) for neutrino μ-like, 95.5 ± 4.1 (syst) for antineutrino μ-like, 73.8 ± 7.1 (syst) for neutrino e-like, 11.8 ± 0.9 (syst) for antineutrino e-like and 6.9 ± 1.3 (syst) for neutrino ν_eCC1π^+-like events in Super K and observes respectively 243 for neutrino μ-like, 102 for antineutrino μ-like, 75 for neutrino e-like, 9 for antineutrino e-like and 15 for neutrino ν_eCC1π^+-like events.

Both a rate only and a rate+shape analysis are performed. In both analyses, fits were performed in the joint fit framework, including all samples as data to constraint either background and the hypotheses tested. The number of observed events in the antineutrino mode e-like spectrum is 9 and the measured rate+shape test statistic, $\Delta\chi^2 = \chi^2\left(\beta = 0\right) - \chi^2\left(\beta = 1\right)$ was found to be -1.502. The significance of the result is shown as a p-value. Taking no $\bar{\nu}_e$ appearance as our null hypothesis we find a p-value of 0.209 ± 0.003 for the rate-only analysis and a p-value of 0.215 ± 0.003 for the rate+shape analysis. Taking $\bar{\nu}_e$ appearance according to the best knowledge of the PMNS matrix as our null hypothesis we find a p-value of 0.210 ± 0.003 for the rate-only analysis and a p-value of 0.092 ± 0.003 for the rate+shape analysis. The Bayes factor, given by the rate+shape analysis was found to be 2.12 indicating a poor preference for no $\bar{\nu}_e$ appearance.

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[8] C.Andreopoulos et al., T2K 6.57×10^{20}-POT Joint 3-Flavour Oscillation Analysis T2K-TN-175, http://www.t2k.org/docs/technotes/175

[9] C.Andreopoulos et al., T2K 2.315×10^{20}-POT 3-Flavour Muon-Antineutrino Disappearance Analysis T2K-TN-243, http://www.t2k.org/docs/technotes/243

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Appendices

A. Optimisation of Oscillation Parameter Priors

When generating fake-data toys that are weighted by the Run 1-9c real-data (see section 2), it was found that the amount of CPU time required to generate these toys was $\sim 80\times$ the CPU time required to generate toys weighted by Asimov A fake-data.

The reason for this is that the constraints on the oscillation parameters given by the Run 1-9c data are substantially stronger than the Asimov A constraints [14]. This results in a larger fraction of the oscillation parameters being generated at more unlikely values and therefore the mean probability of accepting each fake-data toy decreased by a factor of 138.

The large increase in CPU time meant that the toy generation process would require an unreasonably large amount of time and the majority of fake-data toys generated would be immediately rejected. In order to increase the efficiency of the fake-data toy generation process, the oscillation parameter priors ranges were reduced from the ranges used to generate marginalization and fake-data toys in the Run 1-9c joint analysis [14], these ranges will be referred to as the *nominal priors* and are shown in the *Range for marginalisation toys* column of Tab 1.

By restricting the range of the each oscillation parameter prior to the region close to the Run 1-9c best-fit point [14], a larger fraction of fake-data toys generated will have a higher likelihood compared to toys generated using the nominal prior ranges (referred to as the *nominal set*), resulting in a reduction in the required CPU time. However, there is a danger that by reducing the prior range, a toy that would have been accepted from the nominal set would no longer be generated and so would be excluded from the new set of accepted toys. This could cause a change in the measured p-values, which would be undesirable.

To quantify this effect on the p-values, the maximum possible change in p-value is calculated as follows. Equations 6 and 7 for the calculation of the p-value can be re-written as

$$\text{p-value} = p = \frac{n}{N} \tag{15}$$

Where n is the number of accepted toys in the tail of interest and N is the total number of accepted toys in the test-statistic distribution. At worst, all toys that were excluded from the nominal set could have been in the tail of interest, but when using the reduced prior, these excluded toys would have been re-thrown (to keep the total number of accepted toys constant). At worst, all of these re-thrown toys would be outside of the tail of interest (or vice-versa). Using equation 15, this gives a maximum possible p-value change, $|\Delta p|_{\text{max}}$ of

$$|\Delta p|_{\text{max}} = |p_{\text{nominal}} - p| = \left| \frac{n_{\text{nominal}}}{N} - \frac{n_{\text{nominal}} \pm n_{\text{excluded}}}{N} \right| = \frac{n_{\text{excluded}}}{N} \tag{16}$$

Therefore, by reducing the prior ranges successively for a number of oscillation parameters and calculating $|\Delta p|_{\text{max}}$ for various potential prior ranges, the CPU time required can be minimised whilst keeping $|\Delta p|_{\text{max}}$ smaller than the minimum error on the p-values observed in the Asimov

A sensitivity study (see section 3), which was 8.0×10^{-4}. This is demonstrated for a number of different $\sin^2\theta_{23}$ prior ranges in Tab 10, from which a new prior of $[0.42, 0.62]$ is chosen. Using the same method with this new prior range, Δm^2_{23} and subsequently δ_{CP} have their prior ranges optimised, the results of which are shown in Tab 11.

$\sin^2\theta_{23}$ Prior range	Toys excluded from the nominal set (of 20,000 accepted toys)	Maximum possible p-value change	CPU time (current / nominal)
$[0.300, 0.700]$ (nominal)	0	0	1
$[0.355, 0.680]$	1.32×10^{-8}	6.64×10^{-13}	0.85
$[0.375, 0.660]$	2.40×10^{-5}	1.20×10^{-9}	0.71
$[0.390, 0.650]$	7.34×10^{-4}	3.97×10^{-8}	0.66
$[0.410, 0.625]$	1.25	6.24×10^{-5}	0.55
$[0.420, 0.620]$	5.04	2.52×10^{-4}	0.52
$[0.430, 0.610]$	52.5	2.62×10^{-3}	no data
$[0.440, 0.605]$	134	6.74×10^{-3}	no data

Table 10: Maximum possible effect on the p-values obtained by varying the $\sin^2 2\theta_{23}$ prior range used to generate toys to fill the test-statistic distributions. *Toys excluded from the nominal set* are the number of toys that would have been accepted (using the MC sampling method) if the nominal prior range had been used but lie outside of the range of the tighter prior under consideration. These are calculated assuming 20,000 accepted toys will be generated.

Parameter	Nominal Prior	Optimised Prior	Maximum possible p-value change (10^{-4})	CPU time (Optimised / Nominal)
$\sin^2\theta_{23}$	$[0.30, 0.70]$	$[0.42, 0.62]$	2.5	0.52
Δm^2_{23}	$[2.30, 2.70]$	$[2.30, 2.65]$	4.6	0.45
δ_{CP}	$[-\pi, \pi]$	$[-\pi, 1.1]$ & $[1.8, \pi]$	7.8	0.39

Table 11: Optimised priors for the oscillation parameters that are randomised according to flat prior probability distributions. The maximum possible effect on the p-values and the expected reduction on CPU time are shown. These are calculated assuming 20,000 accepted toys will be generated.